



Find Duplicate Characters Using Bit Manipulation

Problem

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Leaderboard

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Given a string containing lowercase English letters, find and print all the characters that occur more than once.

You must use bit manipulation to track the characters that have already appeared.

Input Format

The first line contains a string S

Constraints

$1 \leq |S| \leq 10^5$ S contains only lowercase English letters (a-z)

Output Format

Print all duplicate characters in the order of their first repeated occurrence. If there are no duplicates, print No duplicates.

Sample Input 0

```
programming
```

Sample Output 0

```
r g m
```

Sample Input 1

```
greater
```

Sample Output 1

```
r e
```

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Language

Python 3



```
1 # Enter your code here. Read input from STDIN. Print output to STDOUT
2 s=input().lower()
3 mask=0
4 count=[]
5 for ch in s:
```



```
-
-
-
-
6     .... bit=1<<(ord(ch)-ord('a'))
7     .... if mask & bit:
8     ....     count.append(ch)
9     ....     print(ch ,end=" ")
10    .... mask|=bit
11
12
13
```

Line: 1 Col: 1



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